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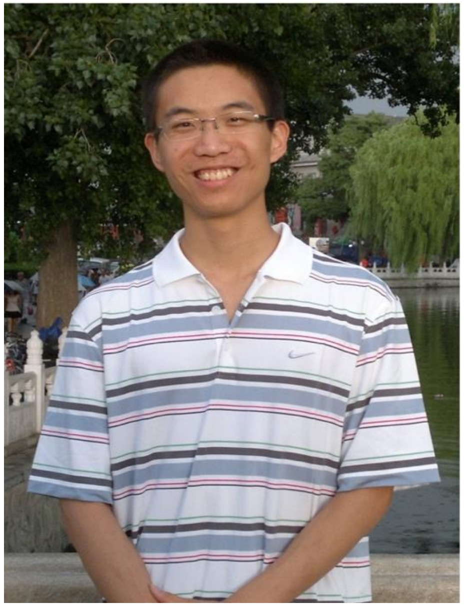
# The Robot Pose and Related Formation Control Problems


Brian D.O. Anderson

Australian National University

- To my major collaborators on this work:



Bomin Jiang

Dr Hatem Hmam

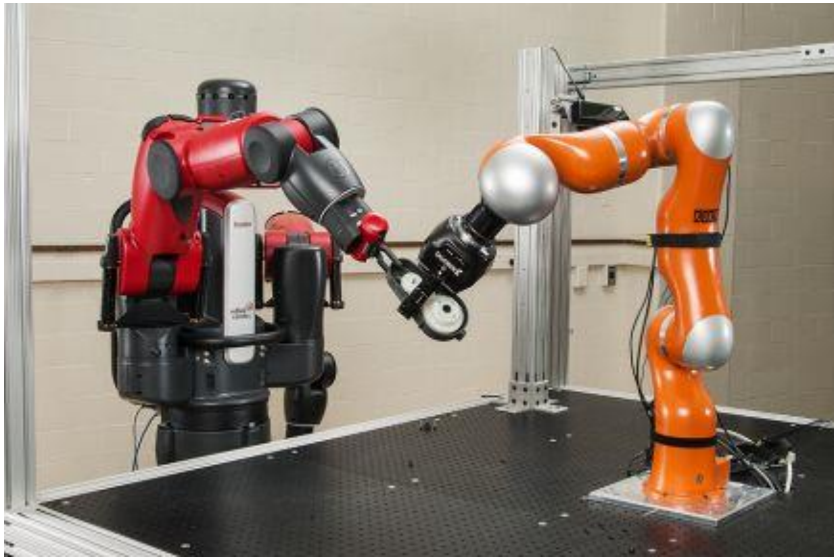
Jiang, B., Anderson, B.D.O. and Hmam, H., **3D relative localization of mobile systems using distance-only measurements via semidefinite optimization,** IEEE Transactions on Aerospace and Electronic Systems, 2019.

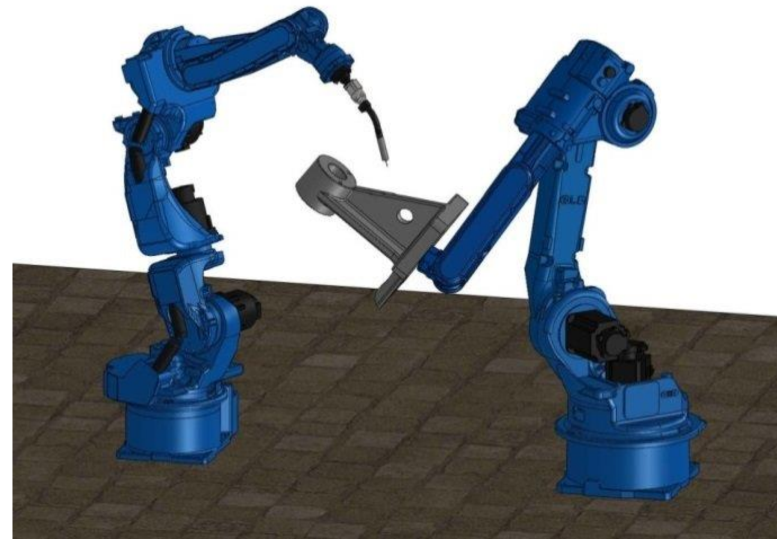
# Outline

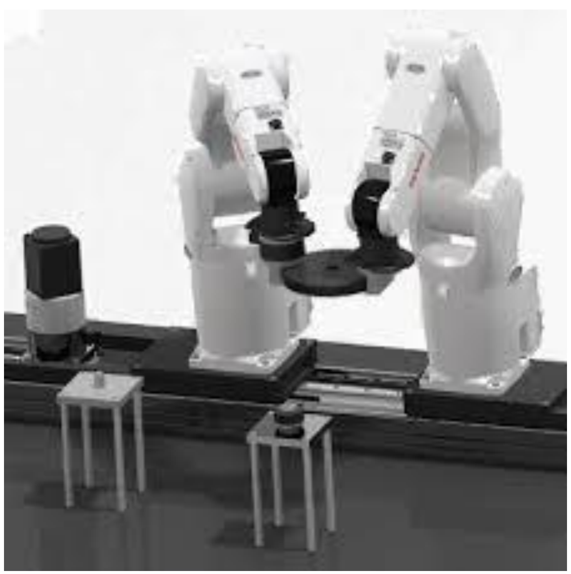
- Introduction & Motivation
  - What is the Robot Pose Problem?
- Related Problems
- Measurement Requirements
- Rotation Matrix Parametrization
- Optimization:
  - Using Semidefinite Programming
  - Using Maximum Likelihood Estimation
- Experiments and Simulations
- Application to formation control
- Conclusions

# Cooperating Robots









# Robot Pose Problem

Two robots must interact cooperatively.

Effective interaction requires knowledge by robot X of the position of the other robot Y, in the same coordinate basis as that of robot X.

Robot Y knows its position in its own coordinate basis, which may not be the same as that of robot X.

**How can inter-robot measurements produce a common coordinate frame, i.e. how can robots localize each other?**

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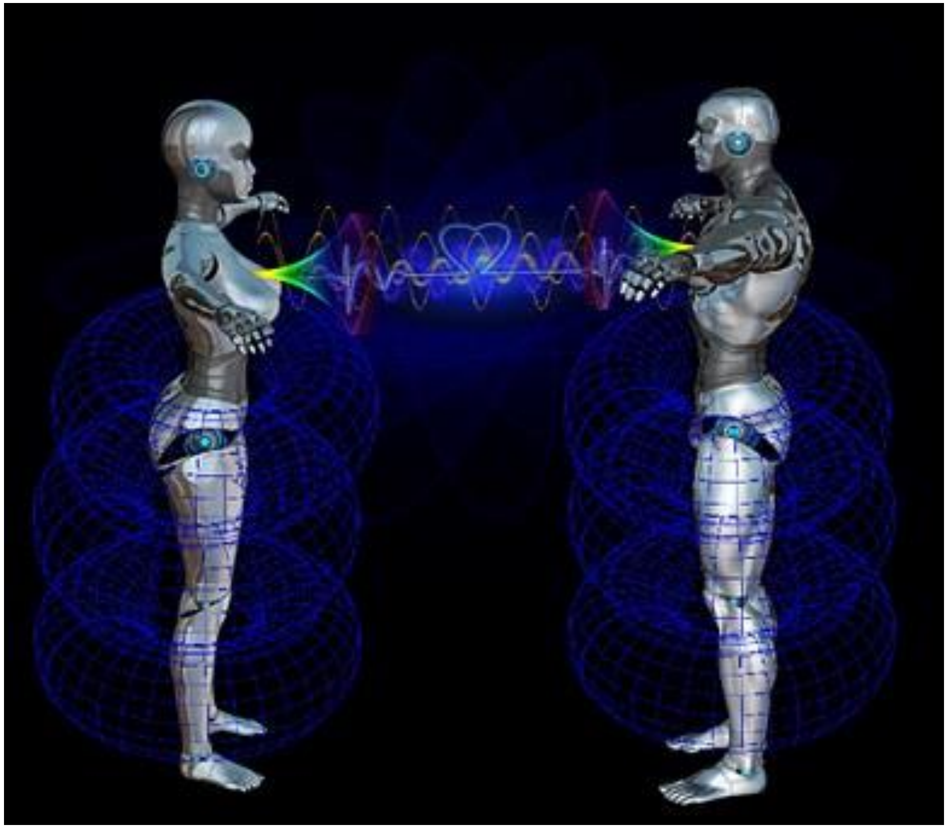
**How can inter-robot measurements produce a common coordinate frame, i.e. how can robots localize each other?**

**This is a special sort of LOCALIZATION problem.**

**It arises in other contexts.**

# Robot Pose Problem

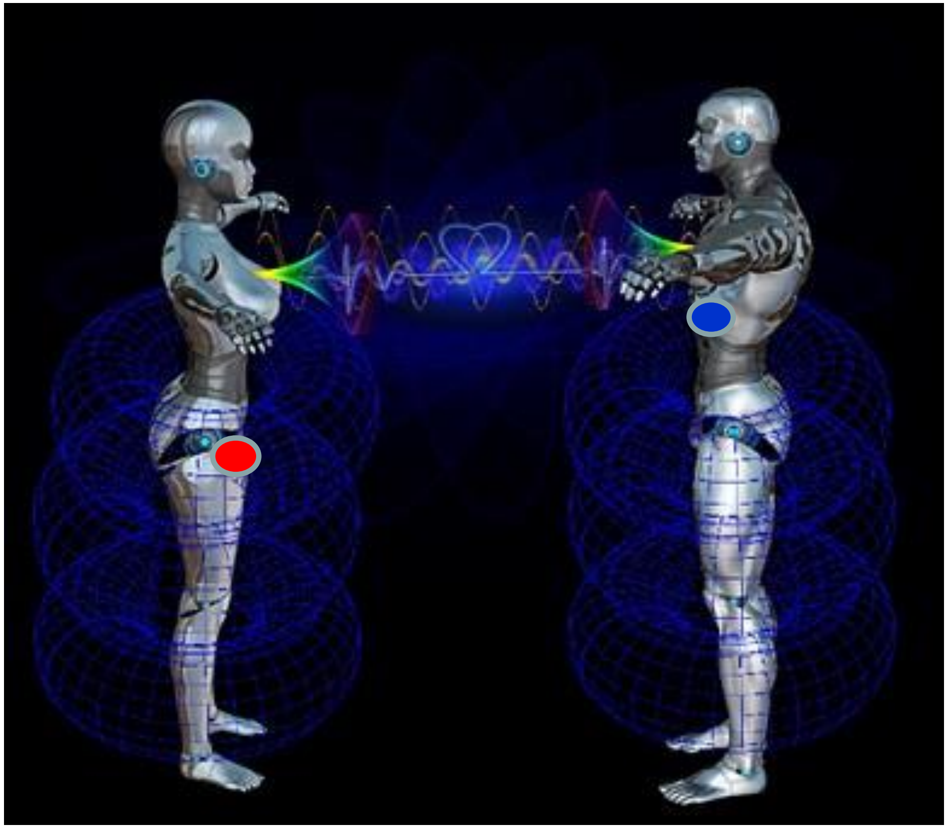


Robot X
Knows Global
Coordinates

Robot Y
Knows Local
Coordinates

# Robot Pose Problem





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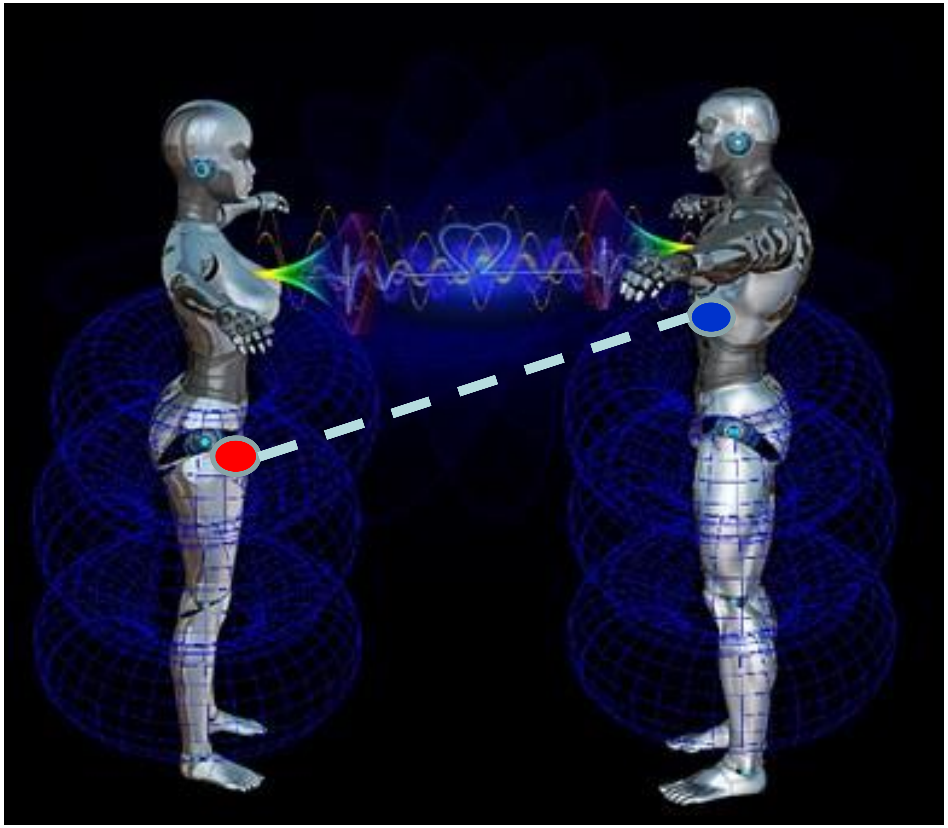
$$p_y = [y_1, y_2, y_3]^\top$$

Local coordinates

$$p_x = [x_1, x_2, x_3]^\top$$

Global coordinates

# Robot Pose Problem



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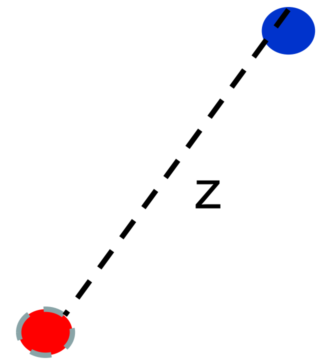


$$p_x = [x_1, x_2, x_3]^\top$$

Global coordinates

# Robot Pose Problem

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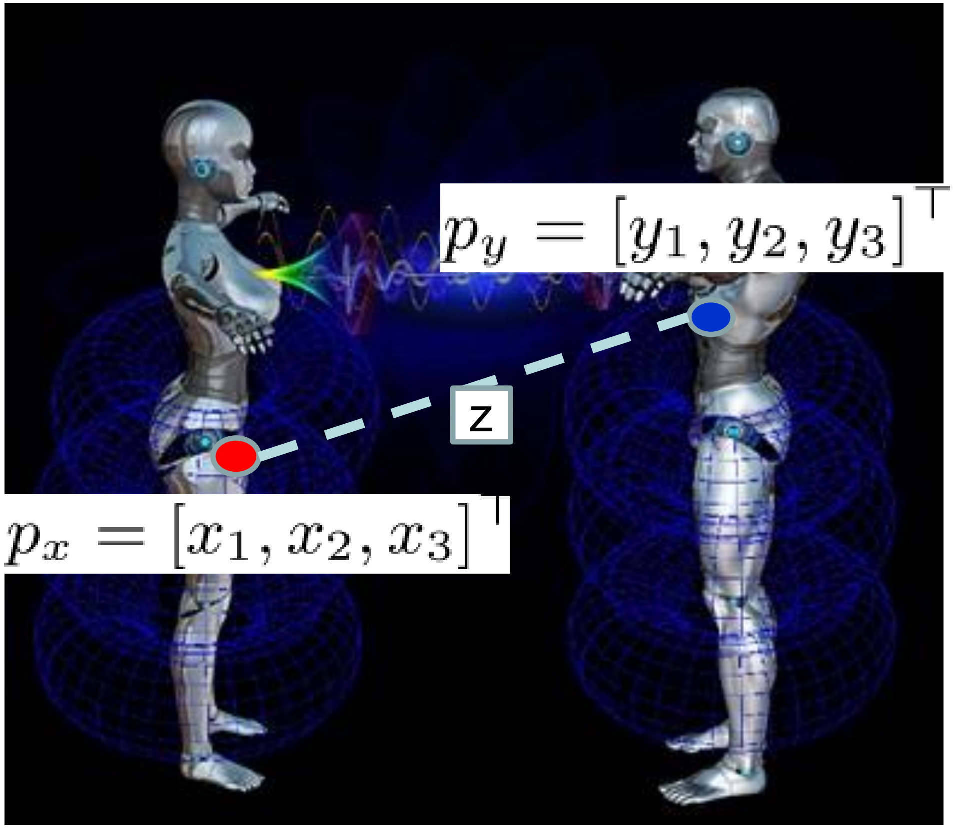


Robot X
Knows Global
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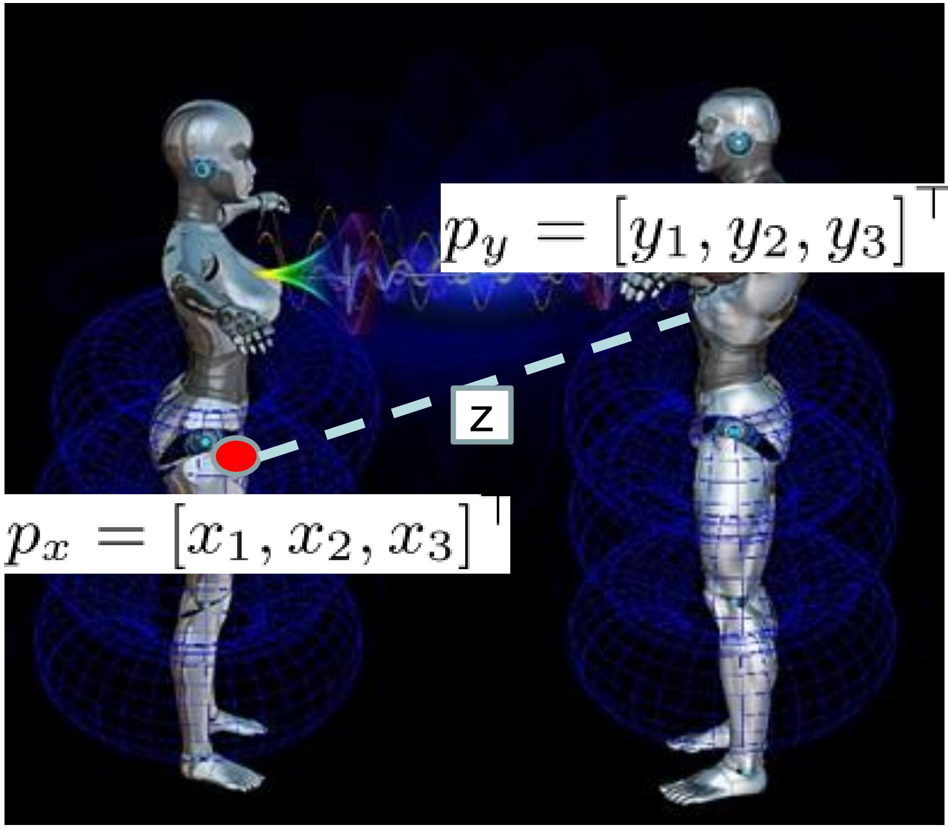
Robot Y
Has Local
Coordinates

*R*: Rotation matrix from global to local coordinates
*T*: Translation vector from global to local coordinates

$$z = ||Rp_y + T - p_x||$$

Coordinates expressed in coordinate frame of Robot X of a point on Robot Y

# Robot Pose Problem

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*R*: Rotation matrix from global to local coordinates
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Knowns

Unknowns

**Key idea: use *multiple* distance measurements to infer *R* and *T***

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- **Related Problems**
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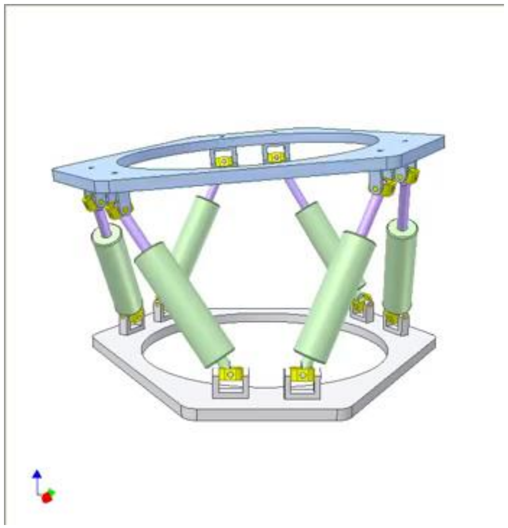
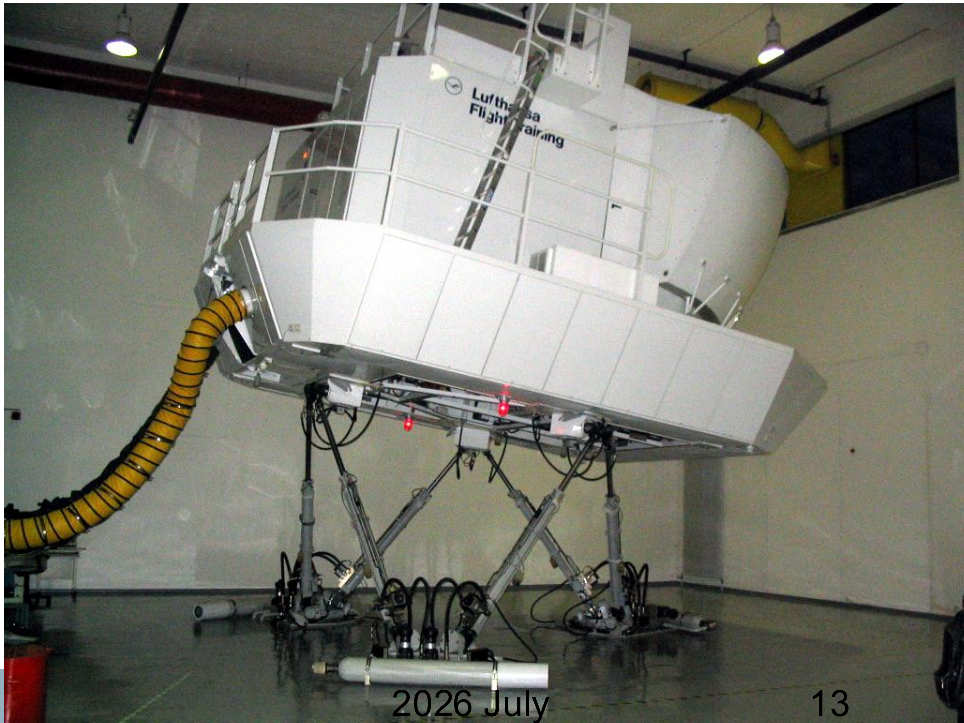
# Related problem: Stewart-Gough platform

- Stewart-Gough platform-1966!
- Two platforms: ground platform, moving platform
- Moving platform has 6 degrees of freedom
- 6 prismatic actuators





# Related problem: Stewart-Gough platform

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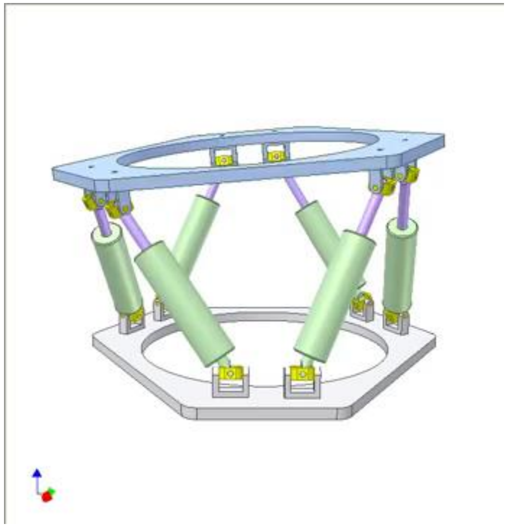
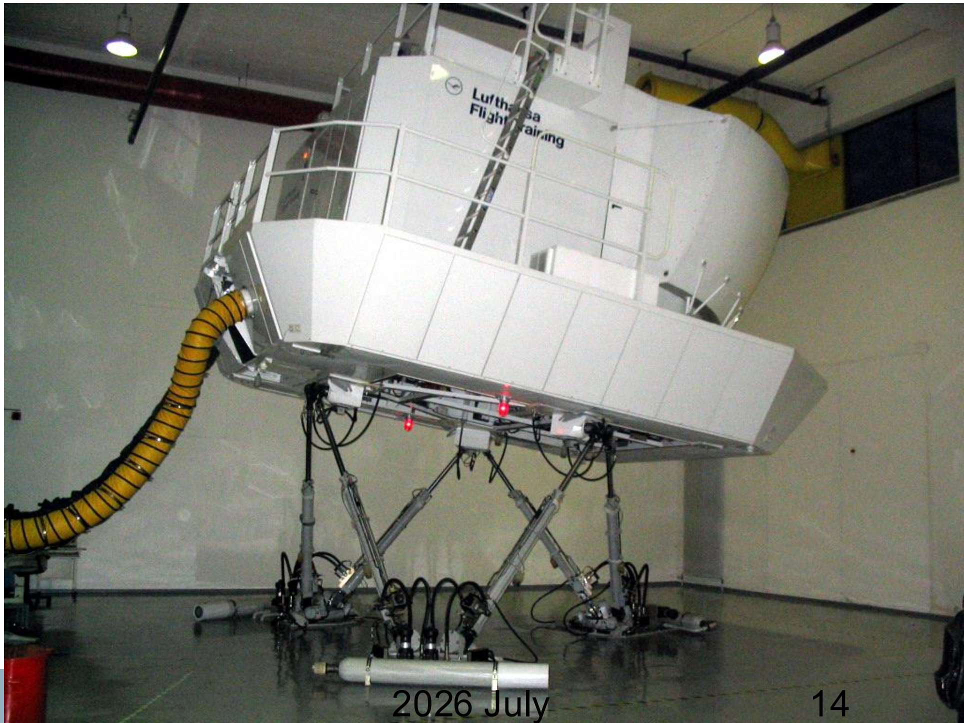
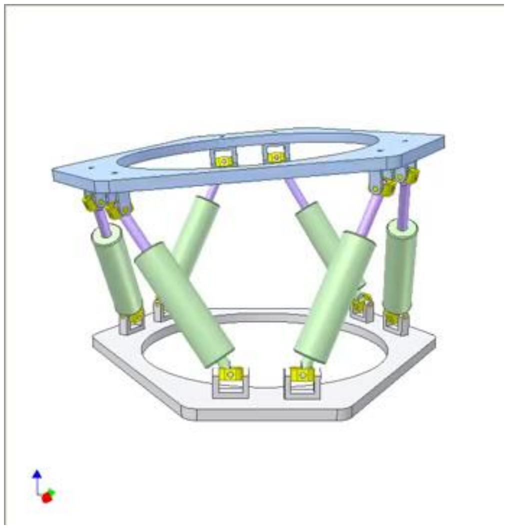
- Stewart-Gough platform-1966!
- 3D localization problem: where is the moving platform?
- Two platforms: ground platform, moving platform
- Two coordinate systems, defined by each platform
- Moving platform has 6 degrees of freedom
- Localization problem has 6 degrees of freedom
- 6 prismatic actuators
- 6 distance measurements

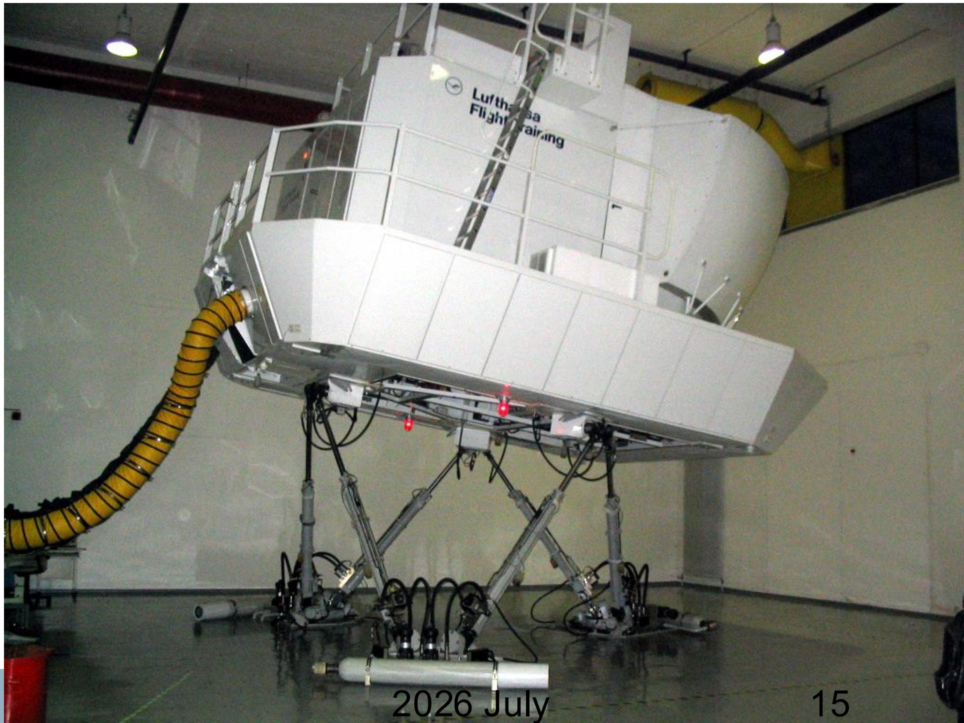




## 6 distance measurements

**CENTRAL QUESTION: How can one obtain orientation and centroid position of upper platform from distance measurements?**





# Related Problem: UAV Localization

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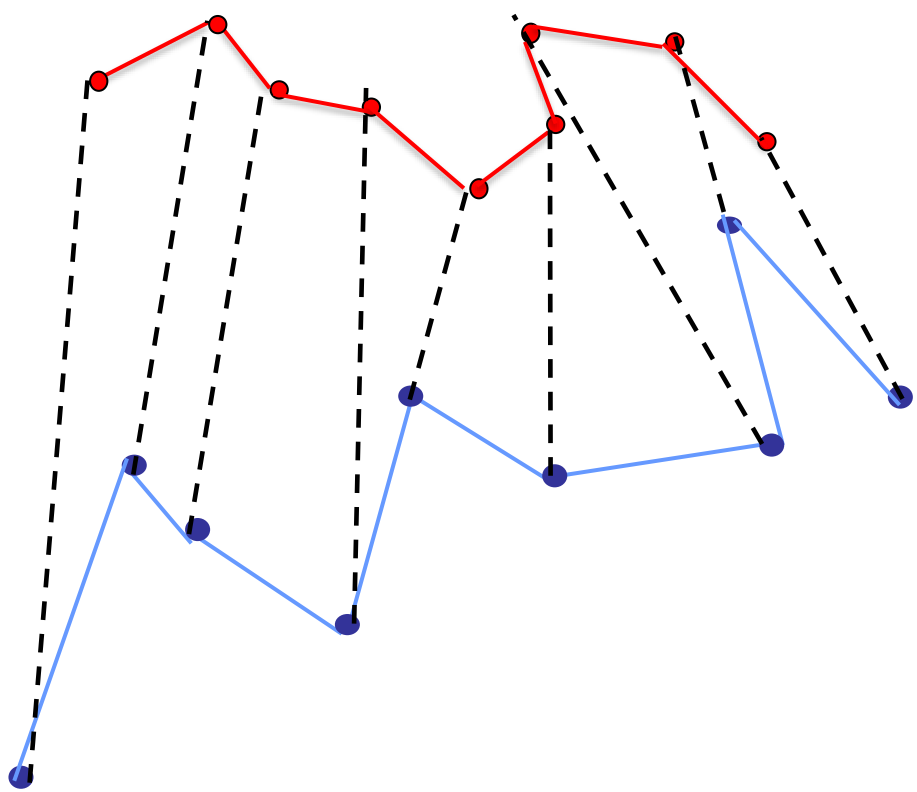
- Two UAVs are flying near one another
- UAV X has GPS (and inertial navigation)
- UAV Y has inertial navigation but is GPS-denied, being in a building canyon or in a building
  - Even with initialised global coordinates, after a few minutes flight, accumulation of drift means global coordinate information is lost.
  - UAV Y knows position in local (INS) coordinates
- Inter-UAV distance measurements are available at particular time instants.
- How may UAV Y be localized?

# 2 UAV localization problem



UAV X tracks positions in global coordinate frame (and can share with Y)

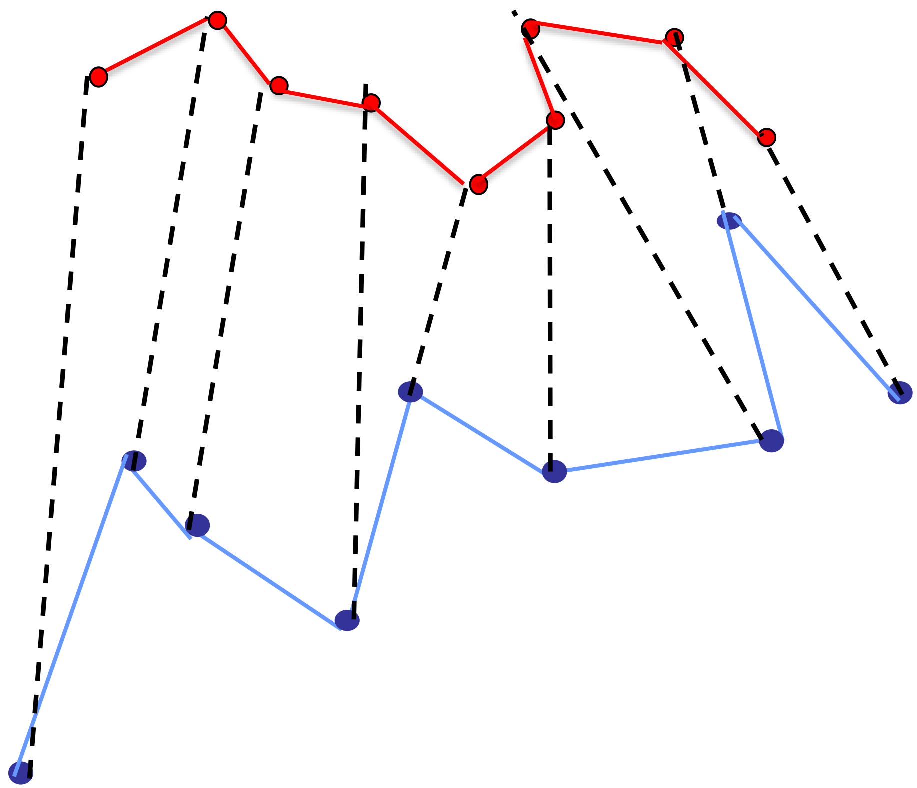
UAV Y tracks positions in local coordinate frame (and shares with X)

Both UAVs know noisy intervehicle distances at discrete time instants.

Where is UAV Y?

# 2 UAV localization problem





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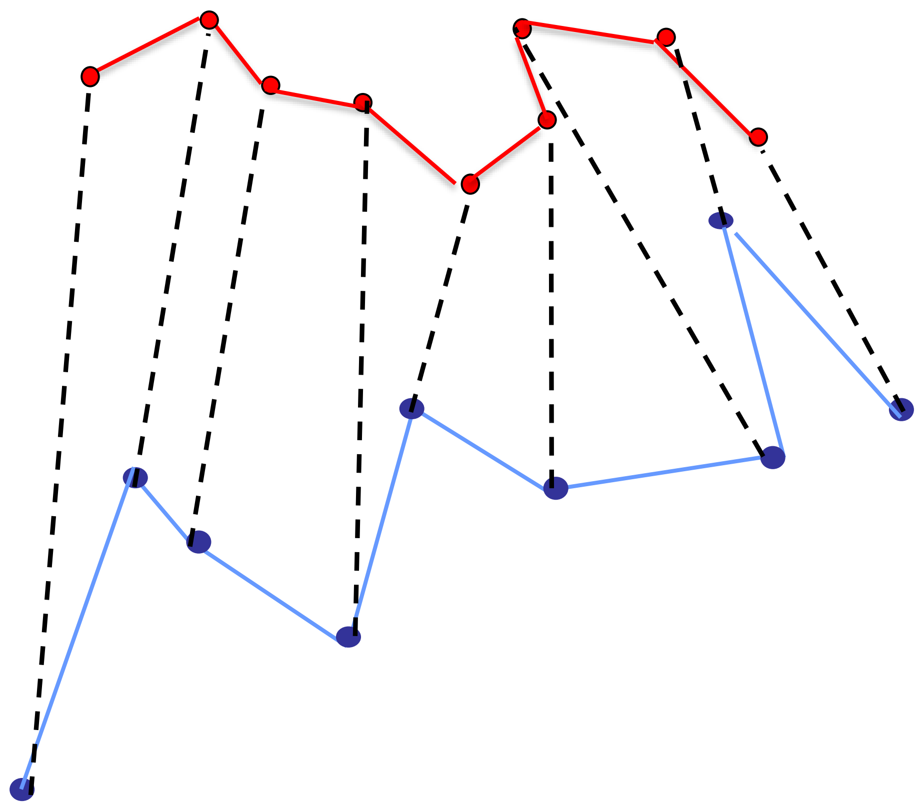

UAV X tracks positions in global coordinate frame (and shares with Y)—***just like points on a rigid body in $R^3$ (like a robot!)***

Where is UAV Y?

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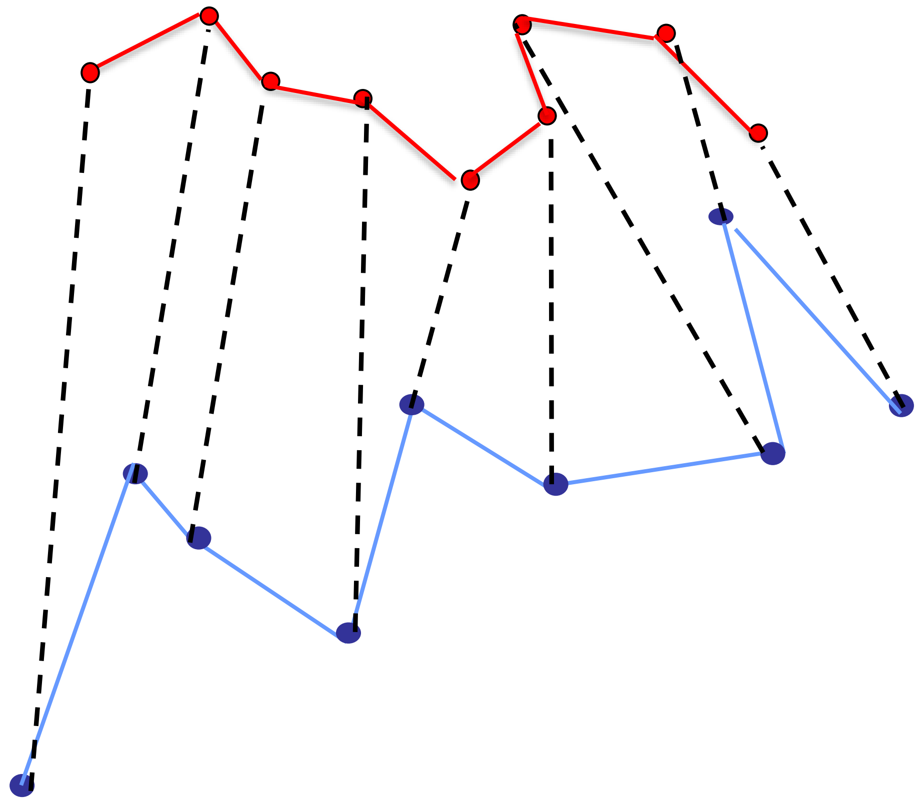

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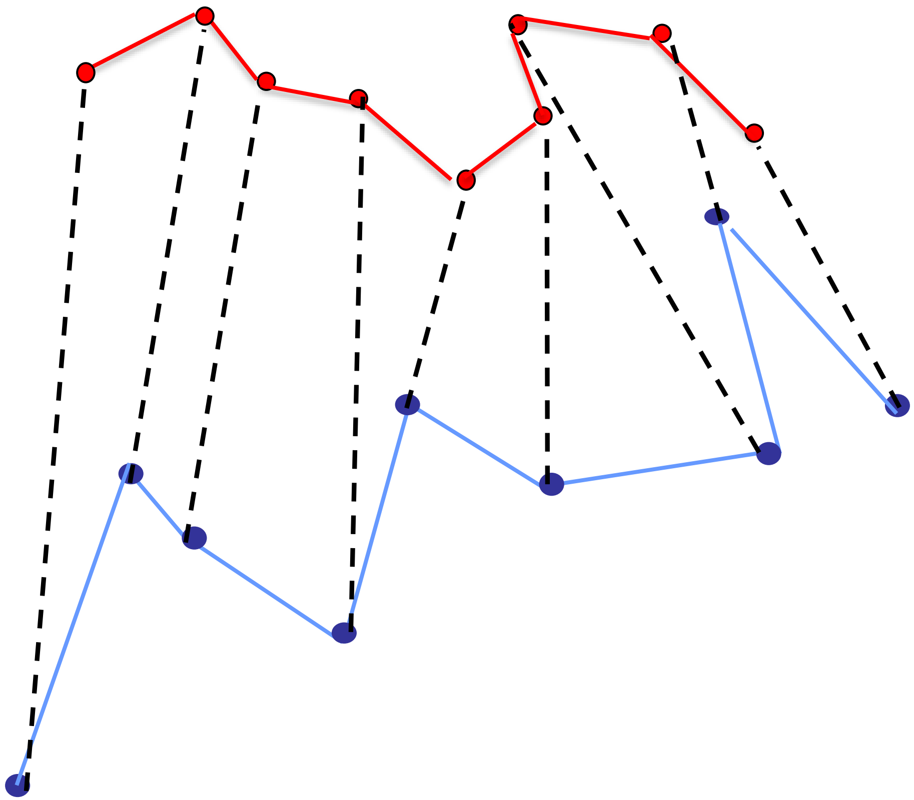

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UAV Y tracks positions in local coordinate frame (and shares with X)—***just like points on a rigid body in $R^3$ (like a robot!)***

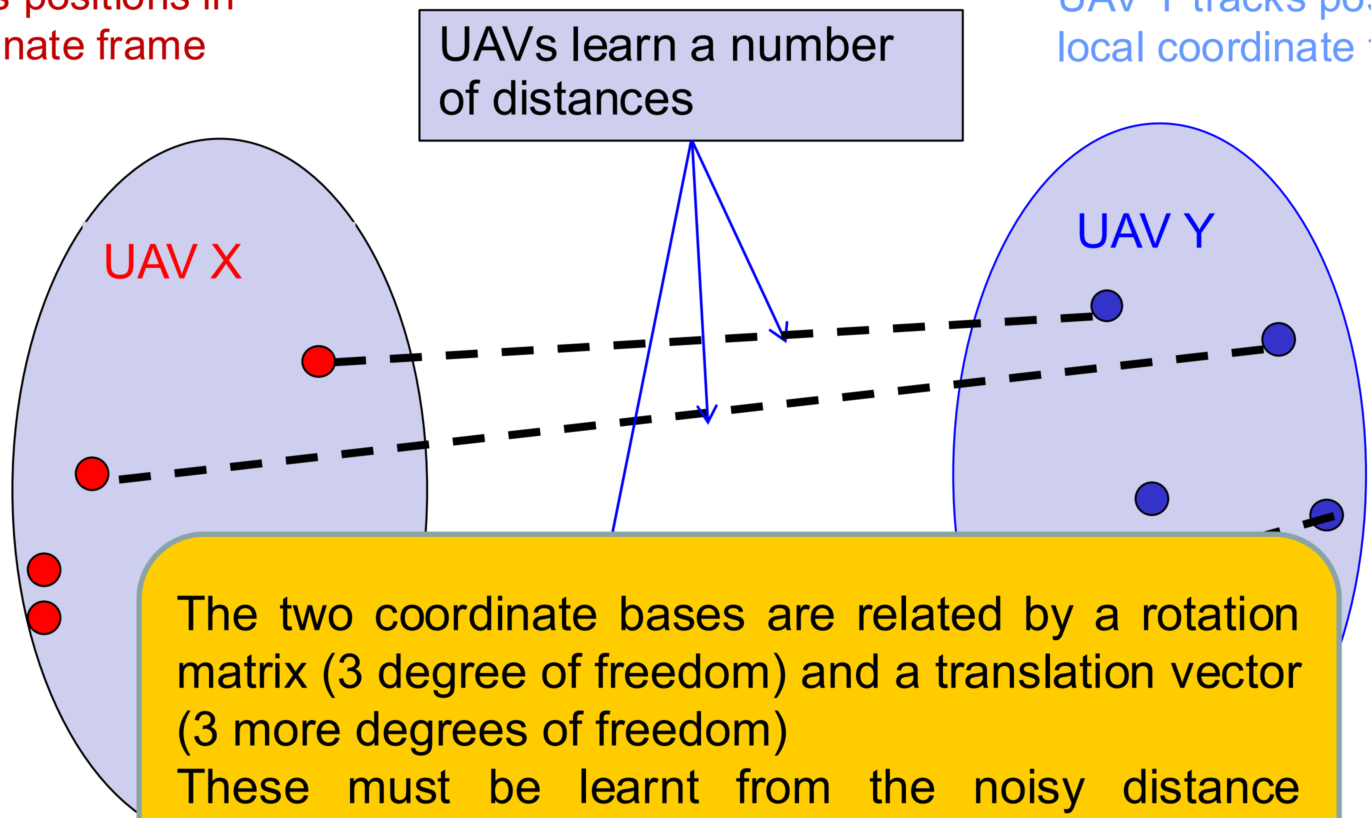
Both agents know **noisy** interagent distances at discrete time instants.

Where is UAV Y?

# 2 UAV localization problem

UAV X tracks positions in global coordinate frame

UAV Y tracks positions in local coordinate frame




The two coordinate bases are related by a rotation matrix (3 degree of freedom) and a translation vector (3 more degrees of freedom)
These must be learnt from the noisy distance measurements.

Robot-like rigid body known in global coordinates

Robot-like rigid body known in local coordinates

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# Robot Pose Measurement Requirements



$$z = \|Rp_y + T - p_x\|$$

Knowns

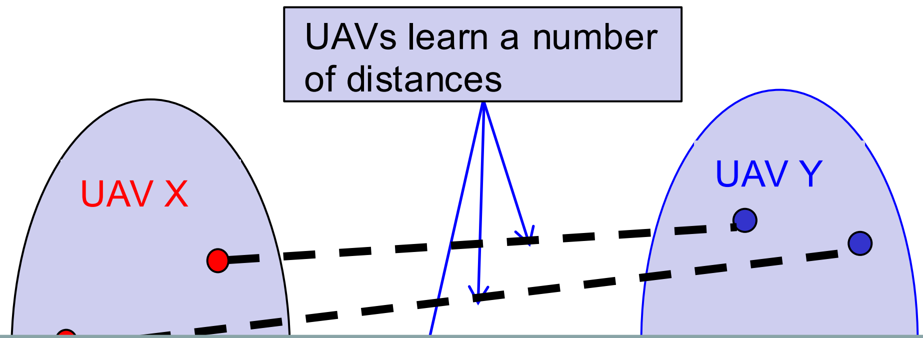
Unknowns

**Key idea: use *multiple* distance measurements to infer *R* and *T***

- There are three degrees of freedom in rotation, and three degrees of freedom in translation
- There are three parameters required to define an orthogonal rotation matrix *R* and three to define a translation vector *T* in three dimensions.
- So six quantities are to be found.
- So we must expect to use *at least* six measurements.

# UAV Localization: Measurement Requirements






The two coordinate bases are related by a rotation matrix (3 degree of freedom) and a translation vector (3 more degrees of freedom)
These must be learnt from the noisy distance measurements.

- Again, there are six scalar quantities to be learnt, three each associated with rotation and with translation.
- A minimum of 6 measurements is clearly needed.
- Since measurements are noisy, more should help.

# Stewart Platform View—decades old

- Finding $R, T$ using **six** distance measurements (actuator lengths) gives rise to **six** polynomial equations.
- Unless all the polynomials are linear, $N$ polynomial equations in $N$ unknowns always have **multiple** solutions (complex and real).
- Successive elimination of five variables for Stewart platform problem yields a **fourtieth** order polynomial equation in one variable.
- Only one solution in general is correct! All solutions may be real.
- To disambiguate, you need further information, e.g. another length (but not an actuator!).

# Stewart Platform View--2

- Same outcome for robot pose problem and two-agent localization: up to 40 solutions with 6 measurements.
- One or more further measurements is required to disambiguate.
  - Generical
  - But becau ter.
- With 7 (or ns, the equation set is over-determined.
- In the presence of noise, the over-determined equation set will have no exact solution.
- Therefore an approximate solution must be sought.

How do you solve a set of overdetermined polynomial equations with noise-induced errors?

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# Rotation Matrix Parametrization

- There are at least three ways to parametrize a 3 × 3 rotation matrix.
- Euler and Tait-Bryan angles. For example, with *X(α)* denoting a rotation of $\alpha$ radians about the x-axis, etc

$$R = X(\alpha)Y(\beta)Z(\gamma)$$

- Quaternions: with $a^2 + b^2 + c^2 + d^2 = 1$

$$R = \begin{bmatrix} a^2+b^2-c^2-d^2 & 2bc-2ad & 2bd+2ac \\ 2bc+2ad & a^2-b^2+c^2-d^2 & 2cd-2ab \\ 2bd-2ac & 2cd+2ab & a^2-b^2-c^2+d^2 \end{bmatrix}$$

- Elementwise Parametrization with **multiple** quadratic constraints, (9 quadratic equations from each of $RR^\top = I, R^\top R = I$ not all independent.)

$$R = \begin{bmatrix} r_{11} & r_{12} & r_{13} \\ r_{21} & r_{22} & r_{23} \\ r_{31} & r_{32} & r_{33} \end{bmatrix}$$

- Remark: Another quadratic constraint forces the determinant to +1:

$$r_{11} = r_{22}r_{33} - r_{23}r_{32}$$

# Rotation Matrix Parametrization-2

- Euler and Tait-Bryan angles:
  - Minimal number of parameters
  - Not polynomial. Both cosine and sine terms appear.
- Quaternions:
  - 4 parameters rather than 3
  - Nonuniqueness issue
  - One polynomial constraint
- Element-wise parameterization:
  - 9 parameters rather than 3
  - Exceptionally simple parametrization
  - Multiple polynomial constraints (6 independent)
- Use of the distance constraint:

$$z = ||Rp_y + T - p_x||$$

  - After squaring, gives a cubic equation in unknowns with quarternions and quadratic equation with unknowns with element-wise parameterization.

# Older papers used Quaternions

- In mid 1990s the Stewart Platform was studied and the possibility of 40 real solutions was found
- The argument used quaternions and other calculations to develop simultaneous polynomial equations using 6 distances.
- In 2007, Trawny et al (in robot pose paper) revisited the above and showed:
  - In principle, with one extra measurement (7 in all), a unique solution existed for a generic problem
  - With **10** measurements, a unique solution could be found using much less complicated algebra.
  - The method could handle limited noise in measurements, but collapsed with more noise.

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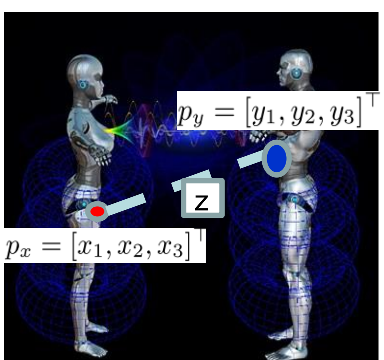
# Setting up an optimization problem

- Suppose:
  - $p_x = [x_1, x_2, x_3]^\top$ is one position on Robot X in the global coordinates. (known)
  - $p_y = [y_1, y_2, y_3]^\top$ is one position on to-be-localized Robot Y in its own coordinate frame. (known)
    - Relation of Robot Y's coordinate frame to Robot X's coordinate frame is unknown

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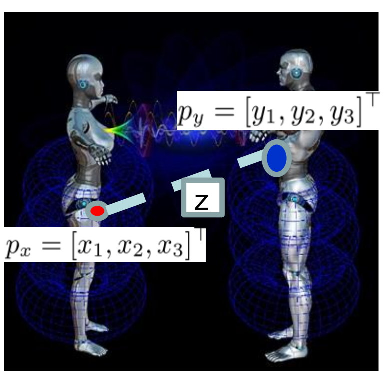


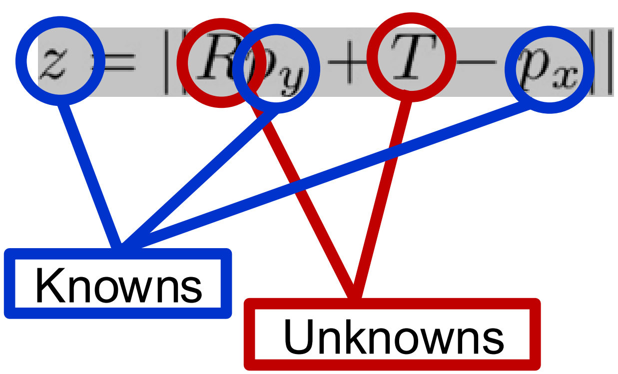

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  - z is one measured distance between the two connected agents:(known)

$$z = \|Rp_y + T - p_x\|$$

- Use **squared version** of this equation. The equation includes **linear terms** in: the $r_{ij}$ (there are 9), the $t_i$ (there are 3) , together with four terms:

$$\sum r_{i1}t_i, \sum r_{i2}t_i, \sum r_{i3}t_i, \text{ and } \sum t_i^2$$

# Setting up an optimization problem

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- With 16 quantities in all, form $\Theta = [\theta_1, \theta_2, \ldots, \theta_{16}]$

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$_2, t_3]^\top$

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$= \|Rp_y$

For the *k*-th measurement $z_k$ we have:

$$A(k)\Theta = b(k)$$

with row vector *A(k)*, and scalar *b(k)* known.

1. The last four entries of $\Theta$ are quadratic functions of the first twelve
2. There are quadratic equations linking the first nine entries of $\Theta$

- Use **squared version** of this equation. The equation includes **linear terms** in: the $r_{ij}$ (there are 9), the $t_i$ (there are 3) , together with four terms:

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- With 16 quantities in all, form $\Theta = [\theta_1, \theta_2, \ldots, \theta_{16}]$

# Setting up an optimization problem

- In summary, squaring all measurement equations, and then collecting them together, yields

$$A\Theta = b$$

where entries of $A,b$ are known and $\Theta$ is the unknown (formed from $R,T$) and has constraints

- There will be no exact solution because of noise.

- We need: **a least squares solution but reflecting the constraints on $\Theta$ entries!**

- **CRUCIAL FA** ... **are all quadratic**

**Choose $\Theta$ to minimize $||A\Theta - b||^2$ subject to some QUADRATIC constraints.**

# Setting up an optimization problem

**Choose $\Theta$ to minimize $||A\Theta - b||^2$ subject to some QUADRATIC constraints.**

This is standard. Semidefinite Programming is the tool. Good software is available.

# Further Properties

- SDP solution method virtually always works, with 7 or more measurements
- With noisy measurements, it is certainly helpful to have more than 7 measurements
- There are “bad geometries”, typically associated with collinearity or coplanarity of points.

# Maximum Likelihood Estimation

- The performance function for SDP is not related to the noise—it is mathematically convenient only.
- If the noise contaminating distance measurements has known probability density, the gold standard solution is the **maximum likelihood estimates** of $R, T$.
- These minimize a different performance index.
- Analytic minimization is impossible. Typically there are multiple minima. **Good initial guess is critical.**
- **Use SDP values to initialize gradient descent ML algorithm.**

# Maximum Likelihood Estimation

- Assumption: Inter-robot distance measurements are contaminated with zero mean gaussian white noise of variance $\sigma^2$. Position data is noiseless.
- Instead of $z = \|Rp_y + T - p_x\|$ there holds

$$\tilde{z}(k) = ||Rp_y + T - p_x|| + \xi, \xi \sim N(0, \sigma^2)$$

- We aim to solve

$$\tilde{R}, \tilde{T} = \arg\min_{R,T} \sum_{k=1}^{N} (\tilde{z}(k) - ||Rp_y(k) + T - p_x(k)||)^2$$

$$\text{subject to } RR^\top = I \;\; detR = 1$$

- We use gradient descent on a manifold, and it always converges to a (local) minimum.

# Gradient Descent on Manifold

$$V = \sum_{k=1}^{N} (\tilde{z}(k) - \|Rp_y(k) + T - p_x(k)\|)^2$$

Manifold is $R^\top R = I$

Usual gradient algorithm:

$$\frac{\partial V}{\partial R} := M(\text{say})$$

Gradient on the manifold:

$$(\frac{\partial V}{\partial R})_{proj} = \frac{1}{2}(M - RM^\top R)$$

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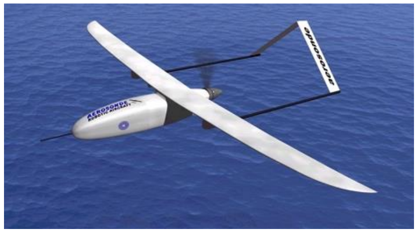
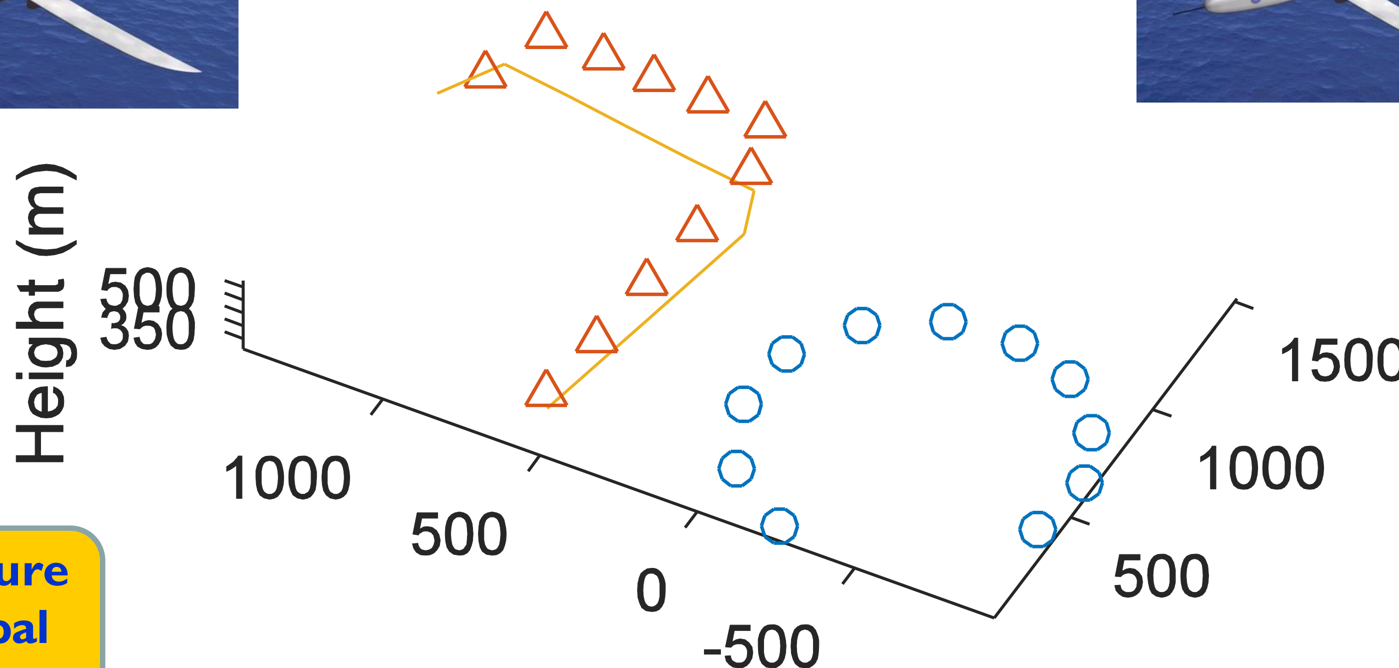
# Experiment flight data for UAV localization






**Entire figure uses global coordinates**

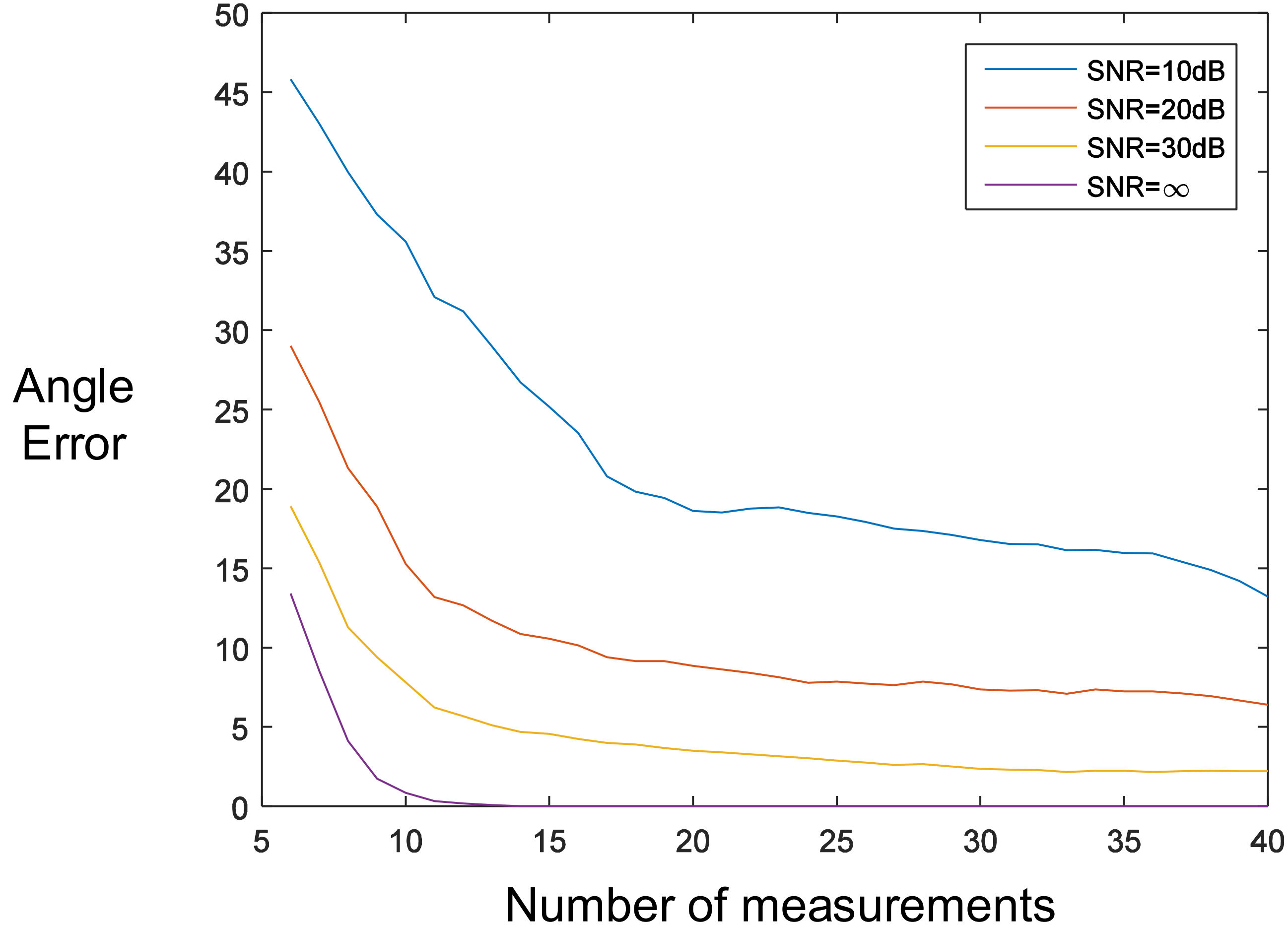
Circles: GPS-equipped UAV X
Triangles: ‘recovered’ positions of GPS-denied UAV Y
(using SDP followed by max likelihood to obtain *R*,*T*)
Solid line: actual trajectory of GPS-denied UAV Y

# Real Trajectories with Simulated Noise

- Further simulations show proposed method works with 10 dB SNR and only 7 measurements.

- Performance shows a balance between number of measurements (quantity of measurements) and SNR (quality of measurements)

Angle error in degrees vs. number of measurements: SNR=inf, 30, 20 and 10



# Outline

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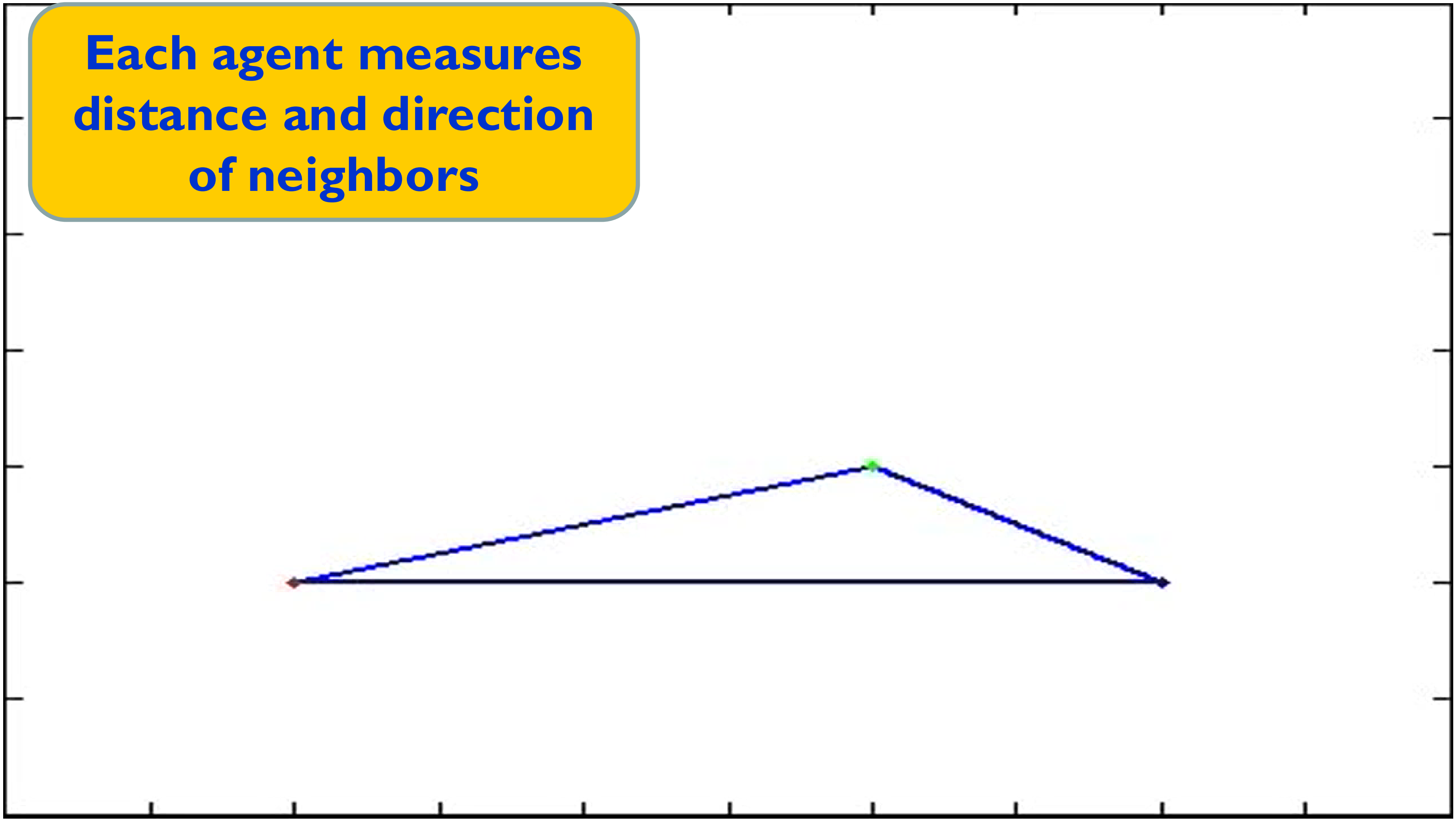
- Introduction & Motivation
  - What is the Robot Pose Problem?
- Related Problems
- Measurement Requirements
- Rotation Matrix Parametrization
- Optimization:
  - Using Semidefinite Programming
  - Using Maximum Likelihood Estimation
- Experiments and Simulations
- Application to formation control
- Conclusions

Consider four problems for multiple UAVs:

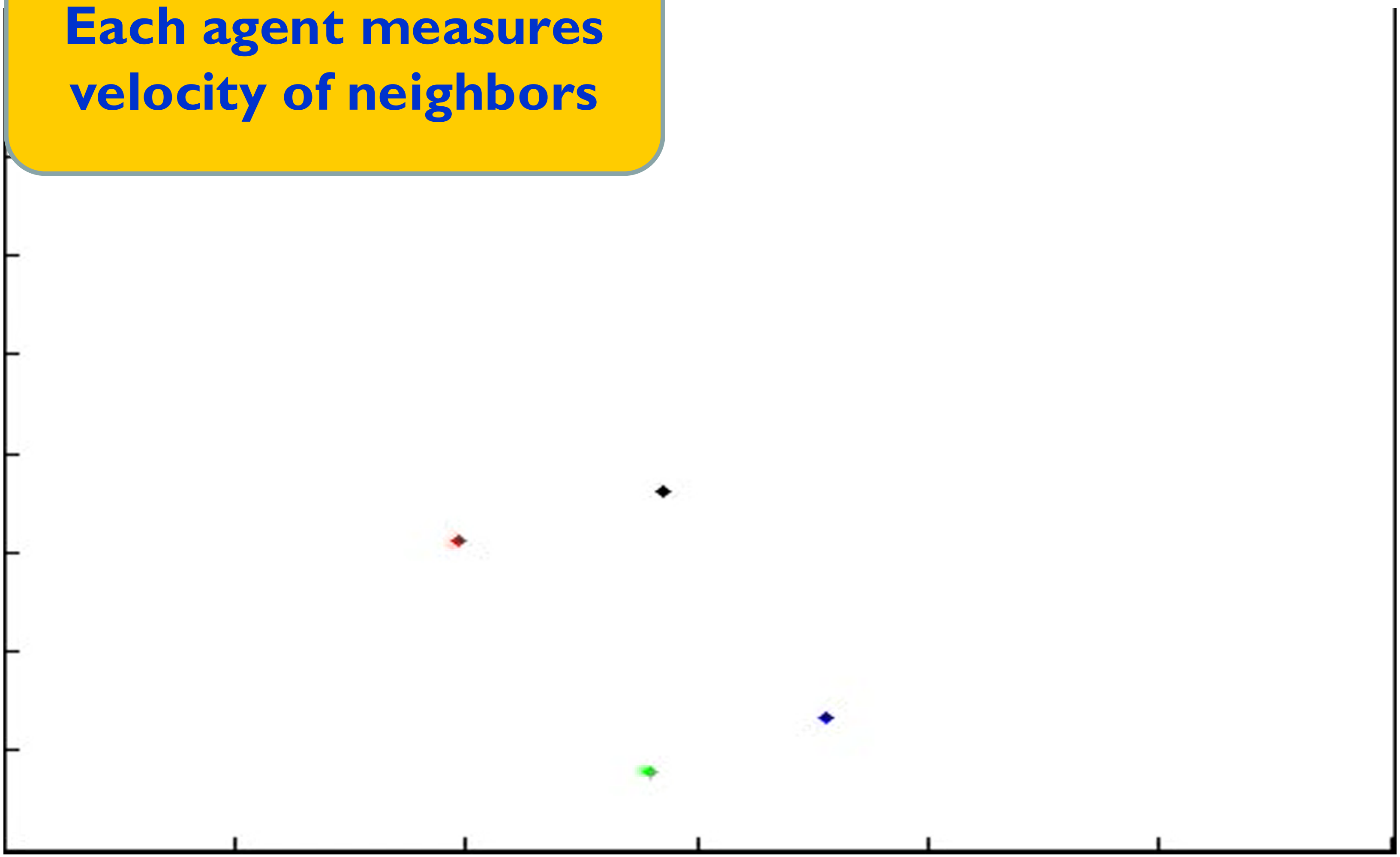
- Form three agents into a triangle of prescribed shape from an arbitrary initial position, using distance and bearing measurements
- Take up velocities, equal to that of a nominated leader agent, using relative velocity measurements.
- Do shape control and velocity consensus simultaneously
- Do shape control and velocity consensus with **only distance measurements** allowed.

# Formation Shape Control : triangle



# Velocity consensus

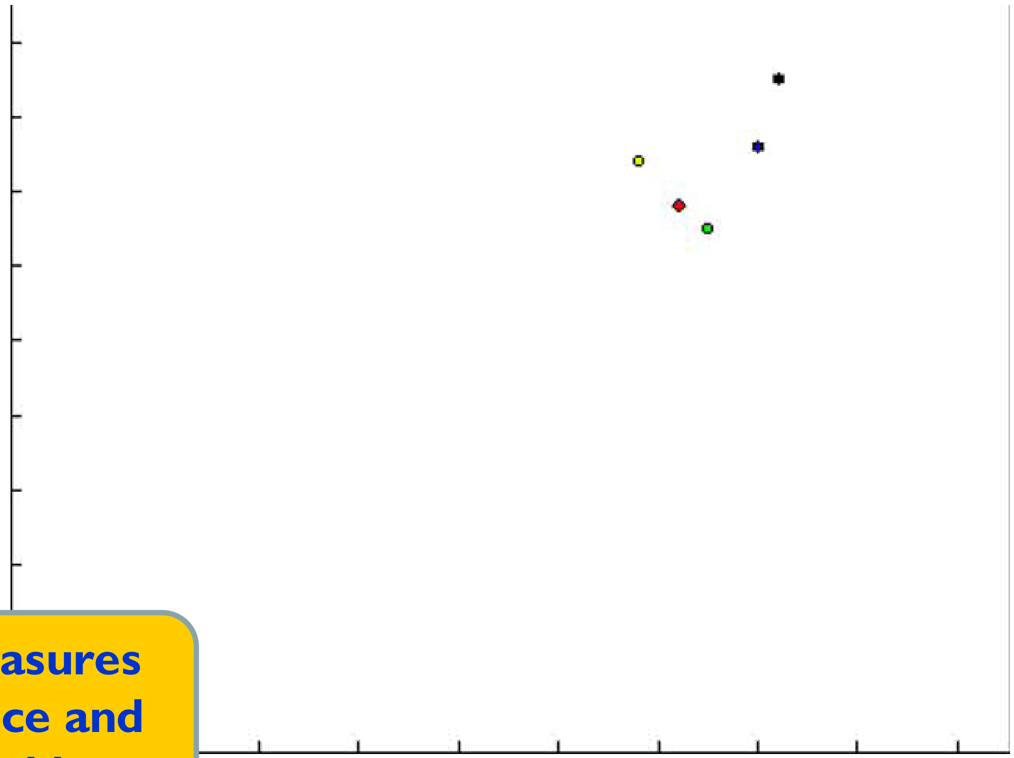
**Each agent measures velocity of neighbors**



# Shape control with flocking

Leader: Blue agent

Objectives:
1. Followers form rectangle, leader in middle
2. Agents do not collide
3. Formation shape preserved in S curve of leader



**Each agent measures velocity, distance and direction of neighbors**

# Measurement-constrained shape and velocity consensus-whole trajectories

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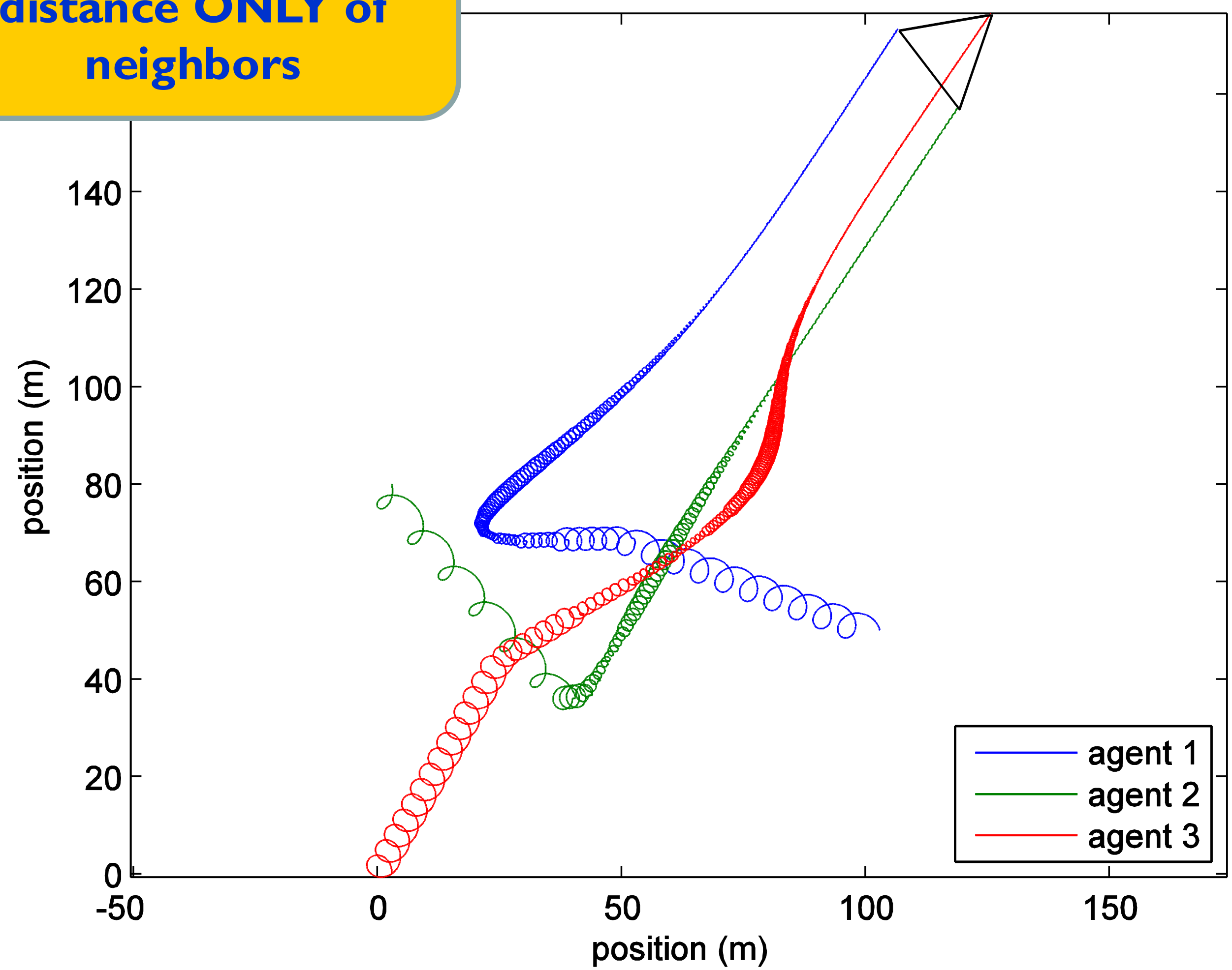
**Each agent measures distance ONLY of neighbors**




Circling is used to localize neighbors with **distance-only** measurements

Circling is wasteful but can be turned off when correct distances are reached

# Measurement-constrained shape and velocity consensus-movie

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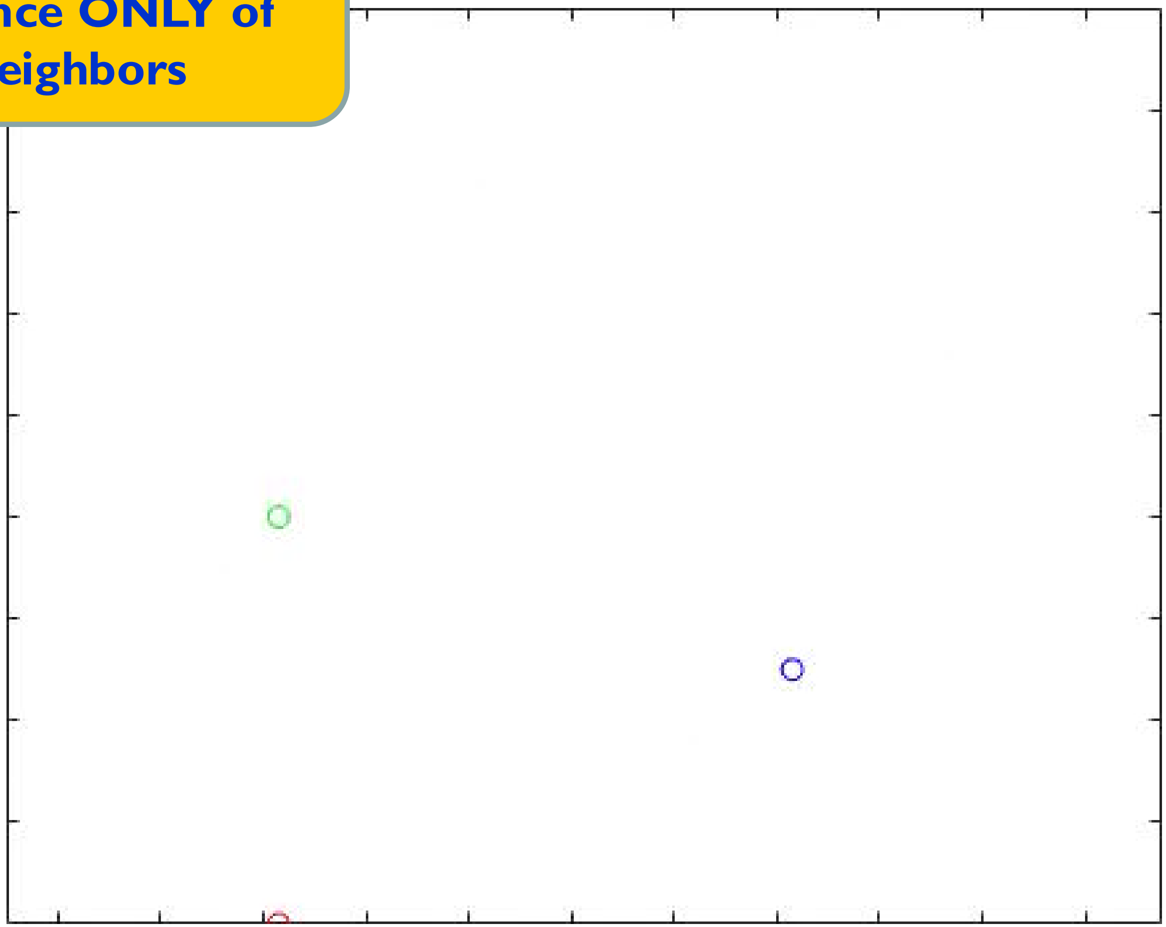
**Each agent measures distance ONLY of neighbors**



Circling is used to localize neighbors with **distance-only** measurements

Circling is wasteful but can be turned off when correct distances are reached

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# Conclusions: Summary of Approach

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- Regard robot pose problem as one of finding a rotation matrix $R$ and translation vector $T$.
- Establish a constrained least squares problem with unknowns including the entries of $R,T$.
- Solve using SDP.
- Use SDP solution to initialize gradient descent algorithm for MLE
  - Gracefully handle gradient descent on a manifold.

# Conclusions: Other Issues

- Multiple robots/UAVs
- Bad geometries
- Offset (bias) errors in distance measurements
- Other types of measurement, e.g. direction.

# Questions